CHAPTER 4

MOVING CHARGES AND MAGNETISM

CHAPTER ESSENCE AT A GLANCE:
Magnetic forces on moving charges, magnetic fields of current elements via Biot-Savart and Ampere's laws, and moving coil galvanometers.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Biot-Savart Principle & Magnetic Fields:** Steady current elements generate surrounding magnetic fields whose strength depends on current magnitude, element length, orientation, and inverse square distance.

• **Ampere's Circuital Law:** Line integral of magnetic field along any closed loop equals the permeability of free space multiplied by total current threading through that loop.

• **Lorentz Force & Helical Motion:** A purely magnetic Lorentz force acts perpendicular to velocity, performing zero work and bending the path into circular or helical trajectories.

• **Force Between Parallel Currents:** Conductors carrying currents in the same direction attract while opposing currents repel, establishing the standard Ampere definition.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$$

Lorentz magnetic force on a moving charge

$$d\mathbf{B} = \frac{\mu_0}{4\pi}\frac{I(d\mathbf{l} \times \hat{\mathbf{r}})}{r^2}$$

Biot-Savart Law for magnetic field of a current element

$$B = \mu_0 n I$$

Magnetic field inside an ideal solenoid

$$F = \frac{\mu_0}{2\pi}\frac{I_1 I_2}{d}L$$

Mutual force between two parallel current-carrying wires

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Radial magnetic field makes θ = 90° so deflecting torque is always maximum (τ = NIAB), producing a perfectly linear scale.

• **Alert:** [Valuation Rubric] Step 1 [Diagram & Geometry]: Draw circular loop in y-z plane with current element dl and axial field point P .

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Biot-Savart law in vector form [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Principle: Biot-Savart Law]** The magnetic field dB produced by a small current element I dl at a position vector r is directly proportional to current I, element length dl, sine of angle θ between dl and r, and inversely proportional to r^2. [1 Mark]

• **[Step 1: Vector Form]** dB = (μ_0 divided by 4π) · [I (dl × r) divided by r^3] = (μ_0 divided by 4π) · [I (dl × unit vector r) divided by r^2]. Here μ_0 divided by 4π = 10^{-7} T m A^{-1} (permeability of free space). Direction of dB is perpendicular to the plane containing dl and r, given by right-hand screw rule. [1 Mark]

• **Key Exam Result:** dB = (μ_0 divided by 4π) · [I (dl × unit vector r) divided by r^2]; Unit: Tesla (T)

2. Ampere's circuital law: statement and formula [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Statement of Ampere's Circuital Law]** The line integral of magnetic field B around any closed loop (Amperian loop) in free space is equal to μ_0 times the total algebraic current enclosed by the loop: closed integral B · dl = μ_0 I enclosed. [1 Mark]

• **[Application: Long Straight Wire]** Amperian circular loop of radius r centered on wire. By symmetry, B is tangential and constant in magnitude: closed integral B · dl = B closed integral dl = B (2π r) = μ_0 I implies B = μ_0 I divided by (2π r). [1 Mark]

• **Key Exam Result:** closed integral B · dl = μ_0 I enclosed; For straight wire: B = μ_0 I divided by (2π r)

3. Magnetic field inside a long straight solenoid [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Setup & Amperian Loop]** Closely wound long solenoid with n turns per unit length carrying current I. Magnetic field inside is uniform and axial; outside field is negligible (B approx. 0). Choose a rectangular Amperian loop abcd of length L and width w: side ab (length L) inside solenoid along axis; cd outside; bc and da perpendicular to axis. [1 Mark]

• **[Step 1: Line ∫ Evaluation]** ■ B · dl = $\int_a^b$ B · dl + $\int_b^c$ B · dl + $\int_c^d$ B · dl + $\int_d^a$ B · dl. 1. Path ab: B || dl ⇒ $\int_a^b$ B dl = B L. 2. Paths bc and da: B ⊥ dl ⇒ ∫ = 0. 3. Path cd: outside solenoid where B ≈ 0 ⇒ ∫ = 0. Total line ∫ = B L. [1 Mark]

• **[Step 2: Enclosed Current & Result]** Total turns enclosed in length L = n L. Enclosed current I encl = (n L) I. By Ampere's Law: B L = μ_0 (n L I) implies B = μ_0 n I. [1 Mark]

• **Key Exam Result:** B = μ_0 n I inside long solenoid; at ends B end = (half (0.5)) μ_0 n I

4. Lorentz magnetic force and trajectory of charged particle in magnetic field [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Lorentz Magnetic Force]** A particle of charge q moving with velocity v in magnetic field B experiences force: F m = q (v × B) = q v B sin θ unit vector n. [1 Mark]

• **[Circular Motion Derivation (θ = 90°)]** When v perpendicular to B, magnetic force acts perpendicular to velocity at every instant, providing required centripetal force without doing work (W = 0, speed v remains constant). q v B = m v^2 divided by r implies Radius r = m v divided by (q B) = p divided by (q B). Period of revolution: T = 2π r divided by v = 2π (m v divided by q B) divided by v = 2π m divided by (q B). Frequency ν = q B divided by (2π m) (orbital gyrofrequency), remarkably independent of speed and radius! [2 Marks]

• **Key Exam Result:** Radius r = m v divided by (q B); Period T = 2π m divided by (q B) (independent of v and r)

5. Force between two parallel current-carrying conductors and definition of 1 Ampere [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Setup & Derivation]** Two long parallel conductors 1 and 2 separated by distance d carrying currents I_1 and I_2 in same direction. Magnetic field produced by wire 1 at position of wire 2: B_1 = μ_0 I_1 divided by (2π d) (directed perpendicular into plane by right-hand thumb rule). [1 Mark]

• **[Step 1: Force on Wire 2]** Force on length L of wire 2 due to field B_1: F_{21} = I_2 L B_1 sin 90° = I_2 L [μ_0 I_1 divided by (2π d)] = (μ_0 I_1 I_2 L) divided by (2π d). Force per unit length: f = F divided by L = (μ_0 I_1 I_2) divided by (2π d). By Fleming's left hand rule, wires attract each other. (Antiparallel currents repel). [1 Mark]

• **[Step 2: Definition of 1 Ampere]** One Ampere is that constant current which, if maintained in two straight parallel conductors of infinite length and negligible cross-section placed 1 meter apart in vacuum, produces between them a force of exactly 2 × 10^{-7} N per meter of length. [1 Mark]

• **Key Exam Result:** f = (μ_0 I_1 I_2) divided by (2π d) N/m; 1 A produces 2 × 10^{-7} N/m at 1 m separation

6. Numerical on force between parallel wires [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Formula]** Force per unit length between parallel currents: f = (μ_0 I_1 I_2) divided by (2π d). [1 Mark]

• **[Step 2: Substitution]** Given I_1 = 10 A, I_2 = 5 A, separation d = 10 cm = 0.10 m, length L = 2 m. f = [($4\pi \times 10^{-7}$) × 10 × 5] divided by [$2\pi \times 0.10$] = ($2 \times 10^{-7} \times 50$) divided by 0.10 = 1.0×10^{-4} N/m. Total force on 2 m length: F = f · L = (1.0×10^{-4} N/m) × 2 m = 2.0×10^{-4} N. If currents flow in opposite directions, force is REPULSIVE. [1 Mark]

• **Key Exam Result:** F = 2.0×10^{-4} N (Repulsive for antiparallel currents)

7. Numerical on ammeter/voltmeter conversion [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Data & Problem]** Galvanometer with G = 12 Ω gives full-scale deflection for I g = 2.5 mA = 2.5×10^{-3} A. (a) Convert to Ammeter of range 0 to 7.5 A: S = [I g divided by (I - I g)] G = [(2.5×10^{-3}) divided by (7.5 - 0.0025)] × 12 approx. (0.0025 divided by 7.4975) × 12 = 4.0×10^{-3} Ω = 0.004 Ω in parallel. [1.5 Marks]

• (b) Convert to Voltmeter of range 0 to 10 V: R = (V divided by I g) - G = [10 divided by (2.5×10^{-3})] - 12 = 4000 - 12 = 3988 Ω in series. [1.5 Marks]

• **Key Exam Result:** (a) Shunt S = 0.004 Ω in parallel | (b) Resistance R = 3988 Ω in series